

Security Incident Report

Incident Title

Potential Brute Force Login Attempt Detected

Incident ID

IR-2026-001

Date/Time Detected

June 3, 2026

Analyst

Dmar Hudson

Severity

Medium

Data Source

Splunk Enterprise, Windows Security Logs, Sysmon

Detection Method

Splunk search query:

index=* EventCode=4625

Executive Summary

A Windows domain account was the target of several unsuccessful authentication attempts. Windows Security Event ID 4625 logs gathered by Splunk were used to identify

the activity. Frequent authentication failures could be a sign of a password guessing or brute force assault on a user's account. .

Affected Systems

- Domain Controller (Windows Server 2022)
- Windows 10 Client
- Active Directory Domain: mydfir.local

Timeline

Time	Event
02:30 AM	Multiple failed login attempts generated
02:31 AM	Event ID 4625 logs forwarded to Splunk
02:32 AM	Reviewed authentication failures
02:35 AM	Investigation confirmed repeated failed logins

Evidence

Event ID: 4625

Source: WinEventLog:Security

Sample Findings:

- Several unsuccessful attempts at authentication
- The same user account is frequently attacked

- Events were successfully imported into Splunk.

Impact Assessment

There was no indication that the account had been successfully compromised. The activity exhibits characteristics of brute force attempts or password guessing. It is advised to continue monitoring.

MITRE ATT&CK Mapping

Technique:

T1110 – Brute Force

Tactic:

Credential Access

Containment Actions

- Examined the activities of the impacted account.
- Confirmed that there were no successful unauthorized logins.
- Kept an eye out for further authentication issues

Recommendations

1. Implement account lockout policies.
2. Enable multifactor authentication where possible.
3. Monitor Event ID 4625 activity using Splunk alerts.
4. Review privileged account usage regularly.

Conclusion

A domain account was the target of numerous unsuccessful login attempts. Using Windows Security logs, Splunk was able to identify the action. During the inquiry, no successful compromise was found.